Sleep, Academic Pressure and Depression Among Students

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***Abstract*—Depression among university students has become an increasing public health concern, with academic pressure and sleep habits recognized as two major factors associated with students' mental health. This study investigated the relationship between sleep duration, academic pressure, and depression using a subset of the Student Depression Dataset. The final analytical sample consisted of 27,901 students. Statistical analyses included a Chi-square test, a Mann–Whitney U test, and logistic regression.**

The results revealed a significant association between sleep duration and depression ($\chi^2 = 276.85$, $df = 4$, $p < 0.001$). In addition, students with depression reported significantly higher academic pressure than students without depression (Mann–Whitney $U = 145,556,846.5$, $p < 0.001$). The interaction between sleep duration and academic pressure was not statistically significant, and a likelihood-ratio test showed that adding the interaction terms did not significantly improve model fit ($LR = 3.31$, $df = 4$, $p = 0.508$).

A simpler main-effects logistic regression model showed that academic pressure was significantly associated with higher odds of depression (OR = 2.27, 95% CI: 2.22–2.32, $p < 0.001$). Sleeping less than five hours was also associated with higher odds of depression (OR = 1.38, 95% CI: 1.28–1.49, $p < 0.001$), whereas sleeping more than eight hours was associated with lower odds of depression (OR = 0.78, 95% CI: 0.72–0.85, $p < 0.001$).

Overall, the results provide no evidence that sleep duration moderates the relationship between academic pressure and depression. However, both academic pressure and sleep duration were independently associated with depression among university students.

I. Introduction

Depression is one of the most common mental health disorders among university students and has become an increasing public health concern worldwide [1]. Academic life exposes students to numerous stressors, including demanding coursework, examinations, financial concerns, and uncertainty about future careers. These factors may negatively affect students' psychological well-being and increase their risk of developing depressive symptoms.

Among the many factors associated with depression, sleep duration has received considerable attention. Adequate sleep is essential for cognitive performance, emotional regulation, and physical health. Previous studies have shown that insufficient or poor-quality sleep is associated with higher levels of stress, anxiety, and depression among university students [2]. At the same time, academic pressure has consistently been identified as one of the strongest predictors of psychological distress and depressive symptoms among university students [3].

Although both sleep duration and academic pressure are known to influence mental health, their combined effect has been less extensively investigated. In particular, it remains unclear whether sleep duration moderates the relationship between academic pressure and depression. Understanding this interaction may help identify protective factors and contribute to the development of interventions aimed at improving students' mental well-being.

In this study, we analyzed a subset of the Student Depression Dataset. The dataset includes demographic, academic, and lifestyle variables such as age, gender, cumulative grade point average (CGPA), sleep duration, academic pressure, and depression status.

The primary research question addressed in this study was: *Does sleep duration moderate the relationship between academic pressure and depression among students?*

To answer this question, we examined the association between sleep duration and depression using a Chi-square test, compared academic pressure between students with and without depression using a Mann–Whitney U test, and evaluated the combined effects of academic pressure, sleep duration, and their interaction using logistic regression. Together, these analyses provide a comprehensive statistical framework for investigating the relationship between sleep habits, academic pressure, and depression among university students.

II. Results

A. Dataset Overview

The statistical analyses were conducted on a final analytical sample of 27,901 students. The dataset included the variables relevant to the research question, namely sleep duration, academic pressure, and depression status.

B. Association Between Sleep Duration and Depression

Our first hypothesis was that sleep duration is associated with depression among university students.

To test this hypothesis, a Chi-square test of independence was performed. The null hypothesis stated that sleep duration and depression status are independent, whereas the alternative hypothesis stated that an association exists between these variables.

The results revealed a statistically significant association between sleep duration and depression ($\chi^2 = 276.85$, $df = 4$, $p < 0.001$). Therefore, the null hypothesis was rejected, indicating that depression rates differ significantly across sleep duration categories.

As shown in Figure 1, the depression rate varied across sleep-duration groups, with the highest rate observed among students sleeping less than five hours.

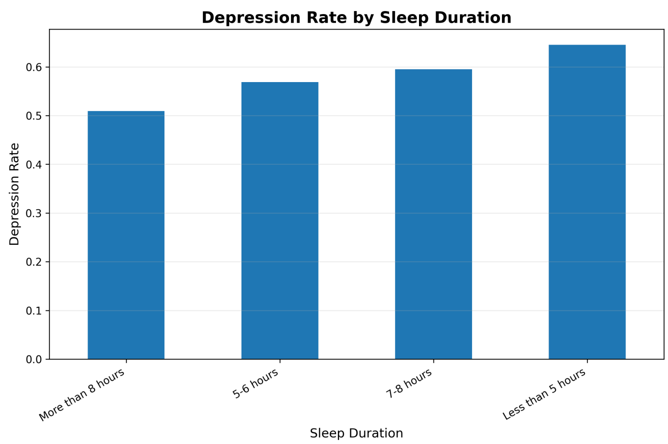

Fig. 1. Depression rate according to sleep duration. Depression rates differ across sleep-duration categories, with the highest rate observed among students sleeping less than five hours.

C. Academic Pressure Among Students With and Without Depression

Our second hypothesis was that students with depression experience higher academic pressure than students without depression.

Since academic pressure is an ordinal variable, a Mann–Whitney U test was used to compare the two independent groups.

The analysis revealed a statistically significant difference in academic pressure between students with and without depression ($U = 145,556,846.5$, $p < 0.001$). The median academic pressure was 4 among students with depression and 2 among students without depression.

These findings indicate that students with depression reported significantly higher levels of academic pressure than students without depression. This difference is also visible in Figure 2.

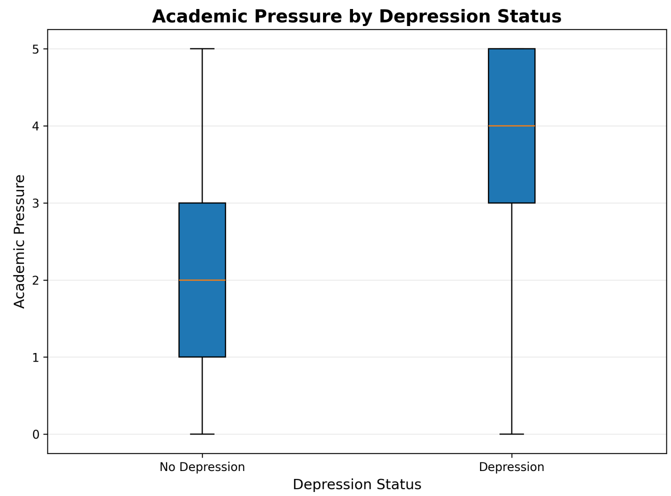

Fig. 2. Distribution of academic pressure according to depression status. Students with depression reported higher academic pressure than students without depression.

D. Logistic Regression Analysis

To address the primary research question, a logistic regression model including interaction terms between academic pressure and sleep duration was first fitted. The interaction terms were not statistically significant. A likelihood-ratio test comparing the interaction model with a model containing only the main effects also showed that the addition of the interaction terms did not significantly improve model fit ($LR = 3.31$, $df = 4$, $p = 0.508$).

These results provide no evidence that sleep duration significantly moderates the relationship between academic pressure and depression. Therefore, a simpler main-effects logistic regression model without interaction terms was fitted to estimate the independent associations of academic pressure and sleep duration with depression.

In the main-effects model, academic pressure was significantly associated with depression (OR = 2.27, 95% CI: 2.22–2.32, $p < 0.001$). Thus, for each one-unit increase in academic pressure, the odds of depression were approximately 2.27 times higher, holding sleep duration constant.

Compared with the reference category of 5–6 hours of sleep, students sleeping 7–8 hours had higher odds of depression (OR = 1.13, 95% CI: 1.05–1.23, $p = 0.002$), and students sleeping less than five hours also had higher odds of depression (OR = 1.38, 95% CI: 1.28–1.49, $p < 0.001$). Students sleeping more than eight hours had lower odds of depression (OR = 0.78, 95% CI: 0.72–0.85, $p < 0.001$). The "Others" sleep-duration category was not statistically significant (OR = 0.68, 95% CI: 0.26–1.83, $p = 0.448$).

Figure 3 illustrates the relationship between academic pressure, sleep duration, and depression. Depression rates increased with academic pressure across all major sleep-duration groups, while the broadly similar trends across groups are consistent with the absence of a statistically significant interaction.

For visualization purposes, the "Others" sleep-duration category and observations with an academic pressure score of 0

TABLE I
MAIN-EFFECTS LOGISTIC REGRESSION RESULTS

Variable	**OR**	**95% CI**	***p*-value**
Academic Pressure	2.27	2.22–2.32	< 0.001
Sleep: 7–8 hours	1.13	1.05–1.23	0.002
Sleep: Less than 5 hours	1.38	1.28–1.49	< 0.001
Sleep: More than 8 hours	0.78	0.72–0.85	< 0.001
Sleep: Others	0.68	0.26–1.83	0.448

were excluded from Figure 3, as these categories contained very few observations.

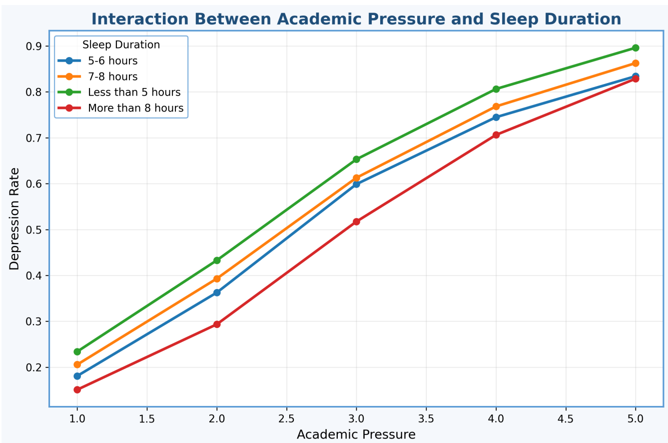

Fig. 3. Relationship between academic pressure, sleep duration, and depression. Depression rates increased with academic pressure across the major sleep-duration groups.

Table I summarizes the odds ratios, 95% confidence intervals, and p-values from the main-effects logistic regression model.

Overall, the logistic regression analyses showed no evidence that sleep duration moderates the relationship between academic pressure and depression. In the simpler main-effects model, both academic pressure and sleep duration were significantly associated with depression.

III. METHODS

A. Dataset

The analyses were performed using the Student Depression Dataset obtained from Kaggle. The Student Depression Dataset contains demographic, academic, and lifestyle information including age, gender, cumulative grade point average (CGPA), sleep duration, academic pressure, and depression status.

The data were imported and processed using Python. Data preprocessing included inspecting the dataset, selecting the variables relevant to the research question, and preparing categorical variables for statistical analysis.

The final analytical sample used in the statistical analyses consisted of 27,901 students.

B. Software and Libraries

All analyses were performed in Python using Jupyter Notebook. The Pandas library was used for data manipulation and preprocessing, NumPy for numerical computations, SciPy for statistical hypothesis testing, Statsmodels for logistic regression, and Matplotlib for data visualization.

C. Chi-Square Test

A Chi-square test of independence was used to examine the association between sleep duration and depression status.

The observed frequencies were compared with the expected frequencies under the assumption of independence. The test statistic, degrees of freedom, and p-value were calculated to determine whether a significant association existed between the two categorical variables.

D. Mann–Whitney U Test

Since academic pressure is measured on an ordinal scale, the Mann–Whitney U test was used to compare academic pressure between students with and without depression.

This non-parametric test compares the distributions of two independent groups without assuming normality. The U statistic and corresponding p-value were computed to evaluate whether the distributions differed significantly.

E. Logistic Regression

Logistic regression was used to model the probability of depression. Depression status was defined as the binary response variable, while academic pressure and sleep duration were included as explanatory variables.

To address the primary research question, an initial logistic regression model included interaction terms between academic pressure and sleep duration to evaluate whether sleep duration moderates the relationship between academic pressure and depression.

The interaction model was compared with a simpler model containing only the main effects of academic pressure and sleep duration using a likelihood-ratio test. Since the interaction terms did not significantly improve model fit, the main-effects model was retained for the interpretation of the individual associations between academic pressure, sleep duration, and depression.

For the main-effects model, coefficients were exponentiated to obtain odds ratios (OR). Odds ratios, 95% confidence intervals, and p-values were reported to assess the magnitude, precision, and statistical significance of the estimated associations.

IV. DISCUSSION

The primary objective of this study was to examine the relationship between sleep duration, academic pressure, and depression among university students, with particular attention to whether sleep duration moderates the relationship between academic pressure and depression.

The Chi-square test revealed a significant association between sleep duration and depression, indicating that depression rates differed across sleep-duration categories. In addition, the Mann–Whitney U test showed that students with depression reported significantly higher levels of academic pressure than students without depression.

The logistic regression analysis did not provide evidence of a moderating effect of sleep duration. The interaction terms between academic pressure and sleep duration were not statistically significant, and the likelihood-ratio test showed that adding these interaction terms did not significantly improve model fit. Therefore, a simpler main-effects model was used to interpret the associations of academic pressure and sleep duration with depression.

In the main-effects model, higher academic pressure was strongly associated with higher odds of depression. Each one-unit increase in academic pressure was associated with approximately 2.27 times higher odds of depression, holding sleep duration constant. Sleep duration was also significantly associated with depression. Compared with the reference sleep-duration category, sleeping less than five hours was associated with higher odds of depression, whereas sleeping more than eight hours was associated with lower odds of depression.

These findings suggest that academic pressure and sleep duration are independently associated with depression among university students, but the present analysis does not provide evidence that sleep duration significantly moderates the relationship between academic pressure and depression.

This study has several limitations. First, the dataset is observational; therefore, causal relationships cannot be established. Second, several variables, including sleep duration and academic pressure, were self-reported, which may introduce reporting bias. In addition, other factors that may influence depression, such as socioeconomic status, family environment, personality traits, and previous mental health conditions, were not included in the analysis.

Future research could investigate additional variables associated with depression, analyze longitudinal data to better understand causal relationships, and evaluate whether interventions aimed at improving sleep habits or reducing academic pressure lead to measurable improvements in students' mental health.

Overall, this study demonstrates the usefulness of statistical methods in identifying factors associated with depression among university students. The findings highlight the importance of considering both lifestyle and academic factors when developing strategies to improve students' psychological well-being.

References

[1] Y. Liu, J. Chen, K. Chen, J. Liu, and W. Wang, "The associations between academic stress and depression among college students: A moderated chain mediation model of negative affect, sleep quality, and social support," *Acta Psychologica*, vol. 239, p. 104014, 2023.

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